



GOA COLLEGE OF ENGINEERING

"BHAUSAHEB BANDODKAR TECHNICAL EDUCATION COMPLEX"
FARMAGUDI, PONDA- GOA - INDIA

MATHEMATICS-III UNIT 1: MATRICES LECTURE 7

DEVELOPED BY MATHEMATICS FACULTY
DEPARTMENT OF SCIENCE & HUMANITIES
GOA COLLEGE OF ENGINEERING



MATRICES LECTURE 8

TOPICS TO BE COVERED:

- Similar Matrices
- Diagonalisation of Matrix
- Minimal Polynomial
- Sylvester's Theorem



Similar Matrices

Two matrices A and B are said to be similar matrices if there exist a non-singular matrix P such that $B = P^{-1}AP$ and it is denoted by $A \sim B$.

$$\text{If } A \sim B \text{ then } B = P^{-1}AP$$

$$PBP^{-1} = A$$

$$\text{If } Q = P^{-1}$$

$$Q^{-1}BQ = P.$$

Two similar matrices have the same Eigenvalues.

$$\begin{aligned} |B - \lambda I| &= |P^{-1}AP - \lambda P^{-1}P| = |P^{-1}(A - \lambda I)P| = |P^{-1}| |A - \lambda I| |P| \\ &= |P|^{-1} |A - \lambda I| |P| = |A - \lambda I| \end{aligned}$$

The characteristic equation of A and B is identical therefore their eigen values are same.



Diagonalization

Reduction of Matrix into Diagonal form

The process of finding a non-singular matrix P such that $P^{-1}AP = D$, where D is diagonal matrix is called **Diagonalization** of A .

Theorem: If a square matrix A order n has n linearly independent eigenvectors, then a matrix P be found such that $P^{-1}AP = D$, where D is diagonal matrix.

Proof: We will prove this result for $n=3$.

Let A be a square matrix of order 3.

Let $\lambda_1, \lambda_2, \lambda_3$ be it's eigenvalues and $X_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$, $X_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$, $X_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix}$ be its corresponding eigenvectors.

Then $A X_1 = \lambda_1 X_1$, $A X_2 = \lambda_2 X_2$ and $A X_3 = \lambda_3 X_3$



$$P = [X_1 \quad X_2 \quad X_3] = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$$

$$\begin{aligned} AP &= A[X_1 \quad X_2 \quad X_3] = [AX_1 \quad AX_2 \quad AX_3] = \begin{bmatrix} \lambda_1 x_1 & \lambda_2 x_2 & \lambda_3 x_3 \\ \lambda_1 y_1 & \lambda_2 y_2 & \lambda_3 y_3 \\ \lambda_1 z_1 & \lambda_2 z_2 & \lambda_3 z_3 \end{bmatrix} \\ &= \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = PD \\ &\quad AP=PD \end{aligned}$$

since the vectors are independent $\mathbf{P}^{-1}$ exists therefore

$$\mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \mathbf{D} \quad \text{Where } \mathbf{D} = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$



Diagonalization

- The matrix which diagonalizes A is called **modal Matrix** of A and it is obtained by grouping eigenvectors of A into a square matrix.
- The resultant diagonal matrix is called the **Spectral matrix** and has the eigenvalues of A as its elements.

- To find A^n :

$$P^{-1}AP = D$$

$$A = PDP^{-1}$$

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD^2P^{-1}$$

$$A^3 = (PDP^{-1})(PDP^{-1})(PDP^{-1}) = PD^3P^{-1}$$

... ..

$$A^m = PD^mP^{-1}$$

$$D^m = \begin{pmatrix} \lambda_1^m & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n^m \end{pmatrix}$$



Examples

1) Find a matrix P which transform the matrix $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ to Diagonal form .

Solution: The characteristic equation of $A = \begin{vmatrix} -2 - \lambda & 2 & -3 \\ 2 & 1 - \lambda & -6 \\ -1 & -2 & -\lambda \end{vmatrix} = 0$

$$\lambda^3 + \lambda^2 - 21\lambda - 45 = 0$$

Eigenvalues are λ : $-3, -3, 5$.

$$\lambda_1 = -3, \lambda_2 = -3 \text{ and } \lambda_3 = 5$$



To find Eigenvectors:

Consider $(A - \lambda I)X = 0$

$$\begin{bmatrix} -2 - \lambda & 2 & -3 \\ 2 & 1 - \lambda & -6 \\ -1 & -2 & -\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ ——— (1)}$$

i) For X_1 put $\lambda = -3$ in equation (1)

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 + 2x_2 - 3x_3 = 0$$

$$2x_1 + 4x_2 - 6x_3 = 0$$

$$-x_1 - 2x_2 + 3x_3 = 0$$

Here we consider only first equation as all equations multiples of each others.

$$x_1 + 2x_2 - 3x_3 = 0$$

Put $x_2 = k$ and $x_3 = l$ then $x_1 = 3l - 2k$



To find Eigenvectors for λ_1, λ_2 :

Put $k = 0, l \neq 0 = 1$ then $X_1 = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$ also put $l = 0, k \neq 0 = 1$ then $X_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$

i) For X_3 put $\lambda = 5$ in equation (1)

$$\begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-7x_1 + 2x_2 - 3x_3 = 0$$

$$2x_1 - 4x_2 - 6x_3 = 0$$

$$-x_1 - 2x_2 - 5x_3 = 0$$

Solve by using Cross-Multiplication Method

$$\frac{x_1}{\begin{vmatrix} 2 & -3 \\ 4 & -6 \end{vmatrix}} = - \frac{x_2}{\begin{vmatrix} -7 & -3 \\ 2 & 6 \end{vmatrix}} = \frac{x_3}{\begin{vmatrix} -7 & 2 \\ 2 & 4 \end{vmatrix}}$$



$$\frac{x_1}{-24} = \frac{x_2}{48} = \frac{x_3}{24} \Rightarrow \frac{x_1}{1} = \frac{x_2}{2} = \frac{x_3}{-1}$$

Therefore the Eigenvector corresponding to eigenvalue $\lambda_3 = 5$ is $X_2 =$

$$\begin{bmatrix} k \\ 2k \\ k \end{bmatrix} \text{ or } \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}.$$

X_1, X_2 and X_3 are linearly independent.

$$P = \begin{bmatrix} 3 & -2 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & -1 \end{bmatrix} \quad P^{-1} = \frac{-1}{8} \begin{bmatrix} -1 & -2 & -5 \\ 2 & -4 & -6 \\ -1 & -2 & 3 \end{bmatrix}$$

$$P^{-1}AP = \frac{-1}{8} \begin{bmatrix} -1 & -2 & -5 \\ 2 & -4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix} \begin{bmatrix} 3 & -2 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & -1 \end{bmatrix} = \begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$



Minimal Polynomial

Let $f(\lambda)$ be a polynomial in λ and A be a square matrix of order n . If $f(A) = 0$, then we say the polynomial $f(\lambda)$ **annihilates (kills)** the matrix A .

A polynomial in x in which the coefficient of the highest power of x is unity is called a **monic polynomial**.

Ex. $x^3 - 3x^2 - 4x + 4$

The characteristic polynomial $|A - \lambda I|$ always annihilates matrix A . (Cayley Hamilton Theorem)

The non-zero monic polynomial of lowest degree which annihilates a matrix is called the **minimal polynomial** of that matrix.

Minimal polynomial denoted by $m(\lambda)$.

Characteristic polynomial $|A - \lambda I|$ is denoted by $f(\lambda)$.



Properties of Minimal Polynomial

- $\deg m(\lambda) \leq \deg f(\lambda)$
- The minimal polynomial $m(\lambda)$ of A is divisor of every $P(\lambda)$, for which $P(A) = 0$.
- Some or all eigenvalues of A satisfy the minimal polynomial equation $m(\lambda) = 0$.
- If minimal polynomial $m(\lambda)$ is identical with the characteristic polynomial $f(\lambda)$, the matrix is said to be non-derogatory.
- If $\deg m(\lambda) < \deg f(\lambda)$, then the matrix is said to be Derogatory.



Method to find minimal polynomial:

1. Consider first degree minimal polynomial $m_1(\lambda) = (\lambda - a)$

If $m_1(A) = 0$, then $m_1(\lambda)$ is the minimal polynomial of A .

i.e., $A - aI = 0 \Rightarrow A = aI$

If $m_1(A) \neq 0$, we proceed to next higher degree.

2. Consider second degree polynomial $m_2(\lambda) = \lambda^2 - a\lambda - b$

If a and b uniquely determined such that $m_2(A) = A^2 - aA - bI = 0$, then $m_2(\lambda)$ is a minimal polynomial otherwise proceed to next step.

3. Consider Third degree monic polynomial $m_3(\lambda) = \lambda^3 - a\lambda^2 - b\lambda - c$

If a, b, c such that $m_3(A) = A^3 - aA^2 - bA - cI = 0$

then $m_3(\lambda)$ is a minimal polynomial, else proceed to next highest degree.

Continue the proceed up to $n - 1$ degree polynomial. If $(n - 1)$ th

degree polynomial is not minimal polynomial then Characteristic polynomial is the Minimal Polynomial.



Examples

1) Find the minimal polynomial of the matrix $A = \begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$.

Solution:

Consider first degree monic polynomial $m_1(\lambda) = (\lambda - a)$

$$m_1(A) = (A - aI) \neq 0 \quad (\text{since } A \neq aI)$$

Therefore $m_1(\lambda)$ is not a minimal polynomial.

Consider second degree monic polynomial $m_2(\lambda) = \lambda^2 - a\lambda - b$

$$m_2(A) = 0 \Rightarrow A^2 - aA - bI = 0 \Rightarrow A^2 = aA + bI$$

$$\begin{bmatrix} 5 & 3 & 1 \\ 4 & 2 & 3 \\ 4 & -2 & 7 \end{bmatrix} = a \begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix} + b \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2a + b & -2a & 3a \\ a & a + b & a \\ a & 3a & -a + b \end{bmatrix}$$



$$\begin{array}{l} 2a + b = 5 \quad a = 4 \quad a = 4 \\ -2a = 3 \quad a + b = 2 \quad 3a = -2 \\ 3a = 1 \quad a = 3 \quad -a + b = 7 \end{array}$$

Since a & b cannot be uniquely determined from above equations, $m_2(\lambda)$ is not a minimal polynomial.

Therefore Characteristic equation of A is the minimal polynomial of A .

$$\text{Now characteristic polynomial of } A = \begin{vmatrix} 2 - \lambda & -2 & 3 \\ 1 & 1 - \lambda & 1 \\ 1 & 3 & -1 - \lambda \end{vmatrix}$$

$$= \lambda^3 - 2\lambda^2 - 5\lambda + 6$$

Therefore the matrix A is non-derogatory.



Sylvester's Method

This is a method to find the matrix $P(A)$ where $P(x)$ is any function.

Let $P(A)$ denote the required function of A .

Assume that $P(A) = c_0A^n + c_1A^{n-1} + c_2A^{n-2} + \dots + c_{n-1}A + c_nI$

Let $f(\lambda) = |\lambda I - A|$

Let $z(\lambda) = \frac{\text{adj}(\lambda I - A)}{f'(\lambda)}$

Then $P(A) = P(\lambda_1)z(\lambda_1) + P(\lambda_2)z(\lambda_2) + \dots + P(\lambda_n)z(\lambda_n)$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are eigenvalues of A .



Examples

1) If $A = \begin{bmatrix} 1 & 1 \\ 4 & -2 \end{bmatrix}$ find e^A .

Solution: Let $P(A) = e^A$

$$f(\lambda) = |\lambda I - A| = \begin{vmatrix} \lambda - 1 & -1 \\ -4 & \lambda + 2 \end{vmatrix} = \lambda^2 + \lambda - 6 = (\lambda + 3)(\lambda - 2)$$

$$f'(\lambda) = 2\lambda + 1$$

Eigenvalues are $\lambda_1 = -3$ and $\lambda_2 = 2$

$$\text{Adj}(\lambda I - A) = \text{adj} \begin{bmatrix} \lambda - 1 & -1 \\ -4 & \lambda + 2 \end{bmatrix} = \begin{bmatrix} \lambda + 2 & 1 \\ 4 & \lambda - 1 \end{bmatrix}$$

$$z(\lambda) = \frac{\text{adj}|\lambda I - A|}{f'(\lambda)} = \frac{1}{2\lambda + 1} \begin{bmatrix} \lambda + 2 & 1 \\ 4 & \lambda - 1 \end{bmatrix}$$

$$P(A) = P(\lambda_1)z(\lambda_1) + P(\lambda_2)z(\lambda_2) = \frac{e^{-3}}{-5} \begin{bmatrix} -1 & 1 \\ 4 & -4 \end{bmatrix} + \frac{e^2}{5} \begin{bmatrix} 4 & 1 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 5.92 & 1.46 \\ 5.87 & 1.51 \end{bmatrix}$$



Alternative method to find A^n , e^A , $\sin A$ and $\cos A$:

Diagonalize A i.e., find non-singular matrix P such that $P^{-1}AP = D$.

i) To find A^n : $A = PDP^{-1}$

$$A^2 = PD^2P^{-1}$$

$$\dots\dots\dots A^n = PD^nP^{-1}$$

$$A^n = P \begin{bmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{bmatrix} P^{-1}$$

ii) To find e^A : $e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$

$$= PP^{-1} + PDP^{-1} + P \frac{D^2}{2!} P^{-1} + P \frac{D^3}{3!} P^{-1} + \dots$$

$$= P \left(I + D + \frac{D^2}{2!} + \frac{D^3}{3!} + \dots \right) P^{-1} = P e^D P^{-1} = P \begin{bmatrix} e^{\lambda_1} & 0 \\ 0 & e^{\lambda_2} \end{bmatrix} P^{-1}$$



iii) To find $\sin A$ and $\cos A$:

$$\begin{aligned}\sin A &= A - \frac{A^3}{3!} + \frac{A^5}{5!} - \frac{A^7}{7!} + \dots \\&= P D P^{-1} - P \frac{D^3}{3!} P^{-1} + P \frac{D^5}{5!} P^{-1} - P \frac{D^7}{7!} P^{-1} + \dots \\&= P \left(D - \frac{D^3}{3!} + \frac{D^5}{5!} - \frac{D^7}{7!} + \dots \right) P^{-1} \\&= P \sin D P^{-1} \\ \sin A &= P \begin{bmatrix} \sin \lambda_1 & 0 \\ 0 & \sin \lambda_2 \end{bmatrix} P^{-1} \\ &\text{and} \\ \cos A &= P \begin{bmatrix} \cos \lambda_1 & 0 \\ 0 & \cos \lambda_2 \end{bmatrix} P^{-1}\end{aligned}$$



Example

- Find the value of $\cos A$ if $A = \begin{bmatrix} 8 & -4 \\ 2 & 2 \end{bmatrix}$.

Solution: The Characteristic equation is $|A - \lambda I| = 0$.

$$\begin{vmatrix} 8 - \lambda & -4 \\ 2 & 2 - \lambda \end{vmatrix} = 0$$
$$\lambda^2 - 10\lambda + 24 = 0$$

Eigenvalues are: $\lambda_1 = 4$ and $\lambda_2 = 6$

Eigenvectors are: $X_1 = \begin{bmatrix} k \\ k \end{bmatrix}$ or $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $X_2 = \begin{bmatrix} k \\ 2k \end{bmatrix}$ or $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Let $P = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ and $P^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$ $D = \begin{bmatrix} 4 & 0 \\ 0 & 6 \end{bmatrix}$

$$\cos A = P \cos D P^{-1} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} \cos 4 & 0 \\ 0 & \cos 6 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} \cos 4 & \cos 6 \\ \cos 4 & 2\cos 6 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2\cos 4 - \cos 6 & -\cos 4 + \cos 6 \\ 2\cos 4 - 2\cos 6 & -\cos 4 + 2\cos 6 \end{bmatrix} = \begin{bmatrix} -2.2674 & 1.6138 \\ -3.2276 & 2.5739 \end{bmatrix}$$

Thank You